

IDATG2208 - Mandatory Assignment - 1

Deadline for submission - 9 Sep, 2026

Dataset

Use the **Superconductivity Dataset** from the UCI Machine Learning Repository (<https://archive.ics.uci.edu/dataset/464/superconductivity+data>). The dataset contains 21,263 superconductors described by 81 features derived from their chemical composition (e.g., atomic mass, valence, ionization energy, thermal conductivity, atomic radius statistics). Your objective is to build a model that predicts the critical temperature at which a material becomes superconducting.

- Features: 81 numerical descriptors computed from the chemical formula of each material (means, weighted means, ranges, and entropies of atomic properties).
- Target variable: `critical_temp` (continuous, in Kelvin).

Note that the target variable is continuous and strongly right-skewed, and that many of the 81 features are highly correlated with one another. Both properties should be taken into account throughout the exercise.

The answers should be submitted through a pdf document on Canvas. The answers should be supplemented with the figures and code snippets. Alternatively, solutions can be provided on Github (link should be provide on Canvas) or take a pdf print from your colab with answers in text boxes of colab.

Each question is mandatory and needs to be answered. A minimum of 70 points is needed for the solution to qualify as valid submission for this mandatory assignment.

1 Exercise-1 [60 Points]

1.1 Data Exploration [4 Points]

- Q1.1.1** Examine the summary statistics (mean, standard deviation, min, max, quartiles) of all 81 features and the target. Which features show the highest variation? Justify why comparing raw standard deviations across these features can be misleading, and propose a more suitable measure.
- Q1.1.2** Plot the distribution of `critical_temp` before and after a log transform. Based on the shape of the distributions, discuss whether the raw or the transformed target should be used for regression later in this assignment.

1.2 Correlation Analysis

[12 Points]

- Q1.2.1** Compute the correlation matrix of all features. Since a full 81×81 heatmap is unreadable, plot a heatmap restricted to the 15 features most correlated with `critical_temp`.
- Q1.2.2** Which variable has the strongest positive correlation with `critical_temp`? Which variable has the strongest negative correlation?
- Q1.2.3** Identify at least 5 pairs of features that are highly correlated with each other (absolute correlation above 0.9), independent of their relationship with the target. What problem does this create for a linear regression model, and how could it be addressed?
- Q1.2.4** Using your correlation analysis, select one feature you expect to be a strong predictor of `critical_temp` and one you expect to be a weak predictor. Justify your choice; you will use these two features in the next section.

1.3 Linear Regression

[12 Points]

- Q1.3.1** Fit a simple linear regression model using gradient descent to predict `critical_temp` using only the weak predictor selected in Q1.2.4. Standardize the feature before running gradient descent and explain why this matters here.
- Q1.3.2** Fit a simple linear regression model predicting `critical_temp` using only the strong predictor selected in Q1.2.4.
- Q1.3.3** Report the regression coefficient and intercept and compare both models.
- Q1.3.4** Plot the regression line against the data points for both features. Does the regression line fit the data well in either case? Why or why not?

1.4 Train-Test Split

[16 Points]

Split the dataset into training (80%) and test (20%) sets in 5 different folds. Train the simple linear regression model (using gradient descent, with standardized features) for each split on the train-test data in each fold. Evaluate the model on the test set in each fold using:

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 score

- Q1.4.1** How well does the strong predictor alone predict `critical_temp` in each split?
- Q1.4.2** How well does the weak predictor alone predict `critical_temp` in each split?
- Q1.4.3** Do you think the model underfits? Why?
- Q1.4.4** Provide the mean and variance from the 5 different folds and comment on the variation in performance across all 5 folds when using the strong predictor versus the weak predictor.

1.5 Multiple Linear Regression

[16 Points]

- Q1.5.1** Train a multiple linear regression model using all 81 features to predict `critical_temp`, using the same splits as in the previous question. Evaluate the model on the test set using MSE, RMSE, and R^2 .
- Q1.5.2** Compare the results of simple vs multiple regression in terms of MSE, RMSE, and R^2 .
- Q1.5.3** Provide comparison plots for multiple versus simple linear regression solved in the previous exercise. At least one of the following plots among (i) Cost vs Iteration (optimization), (ii) Parameter Convergence (coefficients), (iii) Predicted vs Actual (performance), and (iv) Residuals Plot (assumptions check) should be provided.
- Q1.5.4** Which model performs better and why? Given the multicollinearity identified in Q1.2.3, do you trust the individual coefficients of the multiple regression model? Explain.

2 Exercise-2

[40 Points]

- Q2.1** Which features are most suitable/influential in predicting `critical_temp`? (Tip: consider feature importance ranking, and be careful when interpreting importance in the presence of correlated features.)
- Q2.2** The models you trained so far assume a linear relationship between features and target.
- a) **Polynomial regression:** Using a reduced set of the top 10 features from Q2.1, extend the feature space to include quadratic and interaction terms. Does this improve performance? Discuss the risk of overfitting given the increased number of parameters.
 - b) **Regularization:** Train models using Ridge and Lasso regression on all 81 features. Use cross-validation to select the regularization strength α for each. How do these methods affect the coefficients and model generalization? Which features does Lasso eliminate, and does this agree with the multicollinear pairs found in Q1.2.3?
 - c) **Model comparison:** Compare your linear regression results to a non-linear model (e.g., Decision Tree or Random Forest). Which performs better, and why?